Loading and pooling records

CLASP-seq | Capture planning, branch measurements and sequencing inputs

Working document. Red **TO COMPLETE** fields identify missing or conflicting author inputs. Complete the fields affecting a procedure before using it at the bench. Recorded settings are not a claim of experimental optimization.

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01 Use this record with the released protocol

Keep measured quantities, planned inputs and actual inputs separate. Complete one record per experiment and preserve the original count, fragment-analysis and concentration files. Red fields identify decisions or values the method authors still need to supply.

Record	Entry
Experiment / replicate / operator / date	_____
Protocol / construct-manifest / worksheet versions	_____
Raw measurements and sample-sheet location	_____

No loading calculator is supplied yet
The source does not establish the conversion from counted cells or pairs to the custom 10x input, or the calculation basis for the sequencing pool. The tables below preserve the information needed to complete those calculations; they do not prescribe amounts.

02 Map each sample to its capture reaction

Biological sample	10x reaction role	Targeted PCR2 product
Input TCR cells	IN_TCR	IN_TCR
Sorted experimental preparation	EXP_PAIRS	EXP_TCR and EXP_HLA
Input HLA-peptide cells	IN_HLA	IN_HLA

These labels describe roles; assign your own durable sample and channel IDs. The experimental TCR and HLA-peptide products share a capture reaction. Each input reaction has its own barcode namespace. Allocate separate records to additional controls and replicates.

TO COMPLETE - C08
Supply the verified branch-to-index map, including indexed-primer aliases, full sample-index sequences and construct compatibility. Do not infer the released map from the overview's abbreviated primer names.

03 Record counts before calculating capture input

Quantity	Measured value, unit, method and time
Individual cells and viability	_____
Acquired sorter events	_____
Co-positive events and gate denominator	_____
Collected events	_____
Recovered intact pairs and composition check	_____
Material loaded into each reaction	_____
Recovered partitions after analysis	_____

A sorted event is not automatically an intact pair. A recovered partition is not a measurement of cells loaded. Record the units reported by each instrument and retain the measurements used to assess pair composition.

TO COMPLETE - C07
Reconcile the source instructions: 1,000 cells/µL, at least 20 µL available and 'load 20K doublets.' Specify the counting method, cell-versus-pair basis, validated conversion, input limits and target recovery for the custom capture. These source values are unresolved, not executable loading instructions.

Capture record	Planned	Actual
Sample ID / channel	_____	_____
Measured concentration and units	_____	_____
Sample volume and loaded quantity	TO COMPLETE: validated calculation	_____
Target / recovered partitions	TO COMPLETE: target definition	_____
Count-to-loading hold and loading time	_____	_____

Copy this capture record for each reaction. Record the Chromium model, kit/chip lots, firmware and the released custom GEM-X 3′ v4 supplement. Keep the confirmed TSO substitution and other custom steps with the main protocol.

04 Measure the four PCR2 products separately

1. Retain the complete fragment trace and identify the target product using the actual construct sequence and primer coordinates.
2. Record the assay, dilution, target-peak size and concentration with units. Apply a dilution correction once and preserve both the measured and corrected values.
3. Record off-target material and any gel selection. Confirm that each branch meets the released QC criteria before using its concentration to calculate a contribution.

PCR2 branch	Target size / trace ID	Concentration / dilution correction
IN_TCR	_____	_____
EXP_TCR	_____	_____
EXP_HLA	_____	_____
IN_HLA	_____	_____

TO COMPLETE - C10
Supply construct-specific expected sizes, the exact quantification assay and target-peak calculation, cleanup/gel-selection instructions and acceptance criteria. Example sizes in the source are not standards for every construct.

05 Define and document the pool calculation

The source suggests **0.1 IN_TCR : 0.4 EXP_TCR : 0.4 EXP_HLA : 0.1 IN_HLA**. It does not say whether these proportions refer to mass, molar amount or volume.

TO COMPLETE - C10
Confirm the proportion basis and release the calculation with named input units, equations, dilution handling and a checked numerical example. Resolve the source's 'no sample can contribute more than 19 µL' note and specify the total allowable pool volume. Do not assume that either is the final flow-cell loading volume.

PCR2 branch	Desired contribution and units	Actual volume and contribution
IN_TCR	TO COMPLETE: confirmed basis	_____
EXP_TCR	TO COMPLETE: confirmed basis	_____
EXP_HLA	TO COMPLETE: confirmed basis	_____
IN_HLA	TO COMPLETE: confirmed basis	_____
Combined pool	TO COMPLETE: total input	_____

Attach the calculation, its version and the source measurement files. Record any branch that limits the pool and any departure from the confirmed allocation. Retain planned and actual contributions.

06 Keep library preparation separate from flow-cell loading

TO COMPLETE - C11
Pin the SQK-LSK114 amplicon protocol revision, compatible instrument/flow cell, library-preparation input, final loading recipe and required QC. Supply the qualified loading and stopping criteria; the source's loading preference does not establish an optimum.

Stage	Amount, volume, units and supporting record
Pooled amplicons entering library preparation	_____
Recovered prepared library	_____
Prepared library actually loaded on flow cell	_____
Flow-cell QC / run ID / acquisition settings	_____
Remaining material / storage / archive	_____

Archive this record with the sample map, raw QC files and sequencing metadata. The source measurements should be recoverable from the worksheet without reconstructing a calculation from a photograph.

Source notes

Sources are the method authors' laboratory records, as transcribed in the supplied consolidation. The internal source files are not included in this download.

S1. CLASP-seq: the Cell_x_Cell TCR-pMHC pairing protocol. Author-supplied source-checked draft, 11 September 2026. Sections 2.3, 6.1, 6.3-6.4, 7, 10-12. Original laboratory records are cited within that draft; they were not independently reinterpreted for this worksheet.

Protocol and resources